Predicting Benchmark Scores from Sparse and Unpaired Cached Responses

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Abstract

Evaluating a generative model on all queries from all available benchmark tasks is prohibitively expensive. As such, models are often evaluated on different subsets of tasks and/or different subsets of queries from a given task. The resulting collection of responses and the corresponding (model, task) score matrix are unpaired and sparse. In this paper, we introduce the Product Kernel Perspective Space (PKPS), a collection of low-dimensional representations of models based on their embedded responses. PKPS generalizes existing methods by comparing responses to similar, rather than just identical, queries, so models that have responded to different queries are comparable with minimal information lost. Theoretically, we show that benchmark prediction using PKPS is query-efficient relative to methods that only use information from dense and paired sections of the evaluation landscape. Empirically, we show that PKPS outperforms relevant baselines on two tasks—query-efficient evaluation and (model, task) score completion—across two large benchmark suites.

1 Introduction

Evaluating a generative model is necessary for understanding its capabilities. As the capabilities of generative models increase [Wei et al., 2022, Bubeck et al., 2023], their applicability expands to ever more domains—often necessitating the creation of new benchmarks for, e.g., mathematics [Hendrycks et al., 2021], law [Guha et al., 2023], finance [Islam et al., 2023], venture capital [Chen et al., 2025b], agentic tasks [Liu et al., 2024], and even running a vending machine business [Backlund and Petersson, 2025]. The current benchmarking landscape is correspondingly vast: BIG-bench alone contains more than 200 tasks [Srivastava et al., 2023], HELM tracks dozens of scenarios across multiple releases [Liang et al., 2023], and a recent compilation counts 133 benchmarks for frontier models [Zeng and Papailiopoulos, 2026]. As such, a model is rarely evaluated on all benchmarks: of the (model, task) pairs in the compilation above, only 23% have a reported score [Zeng and Papailiopoulos, 2026]. Further, as benchmarks evolve—queries are added, refreshed to avoid contamination [White et al., 2025], or subsampled to reduce cost [Perlitz et al., 2024]—two models evaluated on the same benchmark may have responded to different sets of queries.

Benchmark prediction [Polo et al., 2024, Vivek et al., 2024, Li et al., 2024, Hofmann et al., 2025, Bowyer et al., 2026, Zeng and Papailiopoulos, 2026] attempts to alleviate the resource burden of evaluation by predicting scores from already-available information rather than computing them from newly generated responses. There are two main benchmark prediction tasks: query-efficient evaluation and (model, task) score completion. In query-efficient evaluation, a model responds to a small subset of a task's queries and its score on the full task is predicted from the resulting responses. Existing methods for query-efficient evaluation require access to item-level scores [Polo et al., 2024, Hofmann et al., 2025, Li et al., 2024, Bowyer et al., 2026] and/or cached embedded responses [Helm et al., 2026] to queries that all models have responded to and are typically not applicable to settings where different models have been evaluated on different sets of queries. In (model, task) score completion, a model has not responded to any of a task's queries and its score on the task is predicted from the observed entries of the score matrix. Existing methods for score completion use only the aggregate scores and discard the item-level responses that produced them [Zeng and Papailiopoulos, 2026]. In both tasks, the collection of responses as it is typically available is unpaired and sparse. In this paper, we introduce the Product Kernel Perspective Space (PKPS) to predict benchmark scores

Table 1: MAE of benchmark prediction methods on both suites and tasks (best per column bolded, with ties both bolded; second best underlined; setup described in Section 4.1): query-efficiency at budgets $m \in \{1, 8\}$ queries per cell at full task coverage, and completion of missing cells at cohort sizes $n \in \{10, \max\}$ with task coverage p_{task} and query depth p_{query} fixed at 0.5, except for the sp. columns where p_{task}=0.2 and n=10. "—" denotes a setting where a method is not applicable.

	HELM suite (18 tasks, 93 models)					**EEE suite** (16 tasks, 45 models)				
	Query-eff.		Completion			Query-eff.		Completion		
Method	(m=1)	(m=8)	(n=10)	(n=93)	(sp.)	(m=1)	(m=8)	(n=10)	(n=45)	(sp.)
Sample score	0.292	0.092	—	—	—	0.224	0.081	—	—	—
IRT	0.348	0.343	—	—	—	0.165	0.139	—	—	—
LRMC	—	—	0.139	0.105	0.208	—	—	0.109	0.096	0.144
DKPS	0.168	0.109	—	—	—	0.116	0.077	—	—	—
PKPS	0.136	0.108	**0.120**	0.103	**0.171**	0.076	0.058	**0.082**	0.070	**0.108**
Ensemble	**0.125**	**0.073**	**0.120**	**0.099**	**0.171**	**0.073**	**0.049**	**0.082**	**0.069**	**0.108**

directly from sparse, unpaired collections of responses. Our contributions can be summarized as follows:

1. *Conceptual:* we unify the two main benchmark prediction tasks—query-efficient evaluation and (model, task) score completion—as instances of a single problem: predicting scores from a sparse, unpaired collection of responses from previously evaluated models.
2. *Methodological & theoretical:* we propose the PKPS, a collection of low-dimensional representations of models defined on potentially unpaired embedded responses, and show that benchmark prediction using PKPS is at least as query-efficient as methods that only use information from paired responses (Theorem 2).
3. *Empirical:* on two benchmark suites—93 models on 18 HELM tasks and 45 modern models on 16 tasks from the Every Eval Ever datastore [EvalEval Coalition, 2026]—we show that PKPS matches the accuracy of the sample score with up to an 8× smaller query budget and, when ensembled with low-rank matrix completion, reduces score completion error by up to 28% (Table 1 previews these results; Section 4).

1.1 Related Work

Benchmark prediction. We decompose benchmark prediction into two lines of prior work: query-efficient benchmarking and (model, task) score completion. Methods for query-efficient benchmarking predict a model's score on a full task from its responses to a small subset of the task's queries—typically by selecting an informative query subset and extrapolating via IRT models [Polo et al., 2024, Hofmann et al., 2025], clustering [Vivek et al., 2024], rank-preserving subsampling [Perlitz et al., 2024], learned acquisition policies [Li et al., 2024], regression on item-level scores [Bowyer et al., 2026], or regression on model representations [Helm et al., 2026]. All previous methods proposed for query-efficient benchmarking implicitly require paired responses from previously evaluated models.

Methods for (model, task) score completion predict entries of the score matrix that were never evaluated, via cross-task predictability [Ye et al., 2023], observational scaling laws [Ruan et al., 2024], or low-rank matrix completion [Zeng and Papailiopoulos, 2026]. All operate exclusively on aggregate scores and discard information in the item-level scores or responses. Our proposed method uses item-level information from sparse and unpaired evaluation data to improve on methods from both literatures.

Low-dimensional representations of models. Our approach to sparse and unpaired benchmark prediction first constructs low-dimensional representations of the models before learning a simple regressor in the representation space. Existing approaches to represent models vary by access regime. For example, representations can be induced by comparing internal activations [Duderstadt et al., 2023, Huh et al., 2024] and/or weights [Chen et al., 2025a] in the white-box setting. Evaluators do not necessarily have white-box access and instead rely on black-box methods such as the Data Kernel Perspective Space (DKPS) [Helm et al., 2024], which constructs low-dimensional representations by applying multidimensional scaling to distances between matrices of embedded responses on a paired set of queries. Recent theoretical work has established desirable theoretical properties such as consistency, concentration, and applicability to inference [Acharyya et al., 2024, 2025, Helm et al., 2025]. Most relevant to our setting, Helm et al. [2026] regress benchmark scores of previously evaluated models on DKPS representations and prove that the resulting predictions are query-efficient relative to the sample score. The proposed Product Kernel Perspective Space generalizes this construction with a kernel on the query space, enabling comparisons in the sparse and unpaired evaluation regime.

2 Benchmark Prediction

For our purposes, a model is a random mapping $f : \mathscr{Q} \to \mathscr{X}$ from a query space to a response space. A task is a pair $(Q^{(t)}, s)$, where $Q^{(t)} \subseteq \mathscr{Q}$ is a collection of queries with $M^{(t)} := |Q^{(t)}|$ and s is a scoring function mapping a model and a query collection to a score. Model i's true score on task t is $y_i^{(t)} := s(f_i, Q^{(t)})$.

Evaluation data is indexed by (model, task, query) triples: for n models $f_1, \dots, f_n$ and T tasks $Q^{(1)}, \dots, Q^{(T)}$, the response of model i to query $q \in Q^{(t)}$ may or may not have been generated. Let $Q_i^{(t)} \subseteq Q^{(t)}$ be the queries for which model i's response is observed, with $m_i^{(t)} := |Q_i^{(t)}|$ and $Q_i := \cup_t Q_i^{(t)}$.

The evaluation data is *sparse* when few models have $m_i^{(t)} > 0$ for a given task, and *unpaired* when the cross-model overlaps $m_{ii'}^{(t)} := |Q_i^{(t)} \cap Q_{i'}^{(t)}|$ are small. We let $\rho_{ii'}^{(t)} := m_{ii'}^{(t)} / m_i^{(t)}$ be the ratio of shared queries for a given model pair and note that $\rho_{ii'} \approx 1$ for all pairs in highly curated leaderboards and $\rho_{ii'} \approx 0$ in deployed evaluation settings. The goal of benchmark prediction is to estimate a model's score from the available evaluation data. The two tasks introduced in §1 are the two regimes of the observation count $m_i^{(t)}$:

1. *Query-efficient evaluation* is the case $m_i^{(t)} > 0$. Responses or scores from other models on task t are typically available, and the goal is to outperform the naive sample score $s(f_i, Q_i^{(t)})$.
2. *(Model, task) score completion* is the case $m_i^{(t)} = 0$. No responses from the target (model, task) slice exist and prediction must be based on model i's responses to other tasks and on responses from other models evaluated on task t.

3 The Product Kernel Perspective Space

We seek to estimate a model's score $y_i^{(t)}$ from the sparse, unpaired evaluation data of Section 2. We assume access to the observed responses and their queries, each embedded into a Euclidean space—for $q_j \in Q_i$, a response embedding $x_{ij} \in \mathbb{R}^p$ and a query embedding $u_j \in \mathbb{R}^{p_q}$—and, for the previously evaluated *reference* models, their observed sample scores. We do not require item-level scores, and we do not require that any two models have responded to a common query.

PKPS compares two models by comparing every response of one to every response of the other, weighting each comparison by the similarity of the queries that produced it. Given a *query kernel*

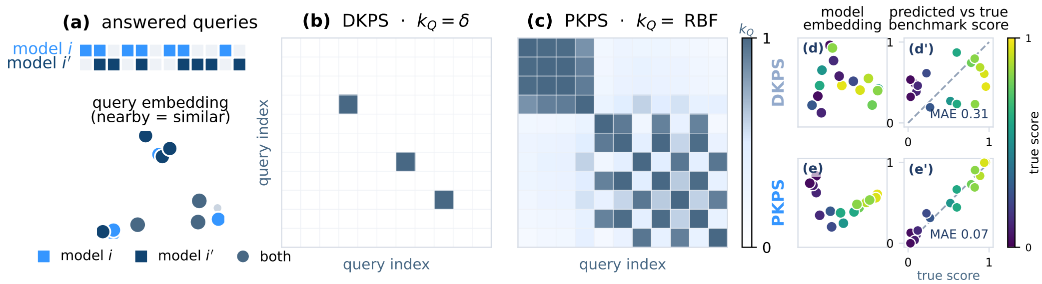

Figure 1: An illustration of the difference between the Product Kernel Perspective Space (PKPS) and the previously proposed Data Kernel Perspective Space (DKPS). **(a)** Top: two models respond to different sets of queries; bottom: queries represented as vectors. **(b)** The query kernel matrix used by DKPS: only information from queries both models have responded to is included in the dissimilarity matrix. **(c)** The query kernel matrix used by PKPS: information is weighted by query similarity in the embedding space. **(d–e′)** A toy benchmark-prediction example. Top: restricted to the rare shared queries, DKPS is noisy and correlates poorly with benchmark score (MAE 0.31). Bottom: PKPS leverages unpaired queries and is highly correlated with score (MAE 0.07).

k_Q and a *response kernel* k_R, the comparison between models i and i' is

$$A_{ii'} = \frac{\sum_{j\in Q_i}\sum_{l\in Q_{i'}} k_Q(q_j,q_l)\,k_R(x_{ij},x_{i'l})}{\sum_{j\in Q_i}\sum_{l\in Q_{i'}} k_Q(q_j,q_l)}, \tag{1}$$

with k_Q evaluated on the query embeddings. Each entry is normalized by its own query-kernel mass, so models that answered different numbers of queries, or entirely different queries, remain comparable. With $A_{ii'}$ in the role of an inner product between the models' response behaviors, the induced squared distance is $D^2(i,i') = A_{ii} + A_{i'i'} - 2A_{ii'}$ (clamped at 0).

This construction strictly generalizes DKPS. With the identity query kernel $k_Q = \delta$, only queries lying in both query sets survive the double sum of Eq. 1, which reduces to the identity-matched comparison $A_{ii'} = \frac{1}{|Q_i\cap Q_{i'}|}\sum_{q\in Q_i\cap Q_{i'}} k_R(x_{iq},x_{i'q})$ of paired DKPS—undefined when models share no queries. For any strictly positive k_Q, such as an RBF over the query embeddings, the normalizer is positive for any nonempty query sets, so Eq. 1 is defined even on disjoint sets and responses to nearby queries contribute in proportion to their similarity (Figure 1). Since the RBF converges pointwise to δ as its bandwidth $\sigma \to 0$, paired DKPS is also the small-bandwidth limit; between the limits, σ controls the tradeoff between exact matching and bridging.

The d-dimensional *representations* are the classical-multidimensional-scaling minimizer

$$(\hat{\psi}_1,\ldots,\hat{\psi}_n) = \underset{z_1,\ldots,z_n\in\mathbb{R}^d}{\arg\min} \sum_{i,i'=1}^{n} \left(\|z_i - z_{i'}\| - D(i,i')\right)^2, \tag{2}$$

so model i is the point $\hat{\psi}_i \in \mathbb{R}^d$, its *perspective*; the collection $\{\hat{\psi}_i\}_{i=1}^n$ is the d-dimensional PKPS. Because Eq. 1 normalizes each pair by its own kernel mass, A is only approximately an inner product and need not be positive semidefinite—hence the clamp on D^2, and classical MDS embeds the doubly-centered matrix by its top-d nonnegative eigenvalues, as is standard. Throughout, we instantiate PKPS with an RBF query kernel and a linear response kernel $k_R(x,x') = \langle x,x'\rangle$ (the synthetic study uses an RBF k_R); linearity makes the kernel factorize across response domains (Section 4.1), and $A_{ii'}$ costs $|Q_i||Q_{i'}|$ kernel terms per pair, with one response-kernel pass serving the whole bandwidth grid.

3.1 Theoretical properties

The representations concentrate around population counterparts at a rate governed by the overlap of the query *distributions* rather than of the query sets. The population PKPS replaces the sums of Eq. 1 with expectations over each model's design distribution (Definition 2, Appendix E). We require bounded kernels and embedded responses, a positive *cross-model query-kernel mass* $\kappa = \inf_{i,i'} \mathbb{E}[k_Q(u,u')]$, and a rank-$d$ spectral gap in the population Gram matrix (Assumptions 1–3, Appendix E). Boundedness and the spectral gap are standard in spectral-embedding analyses; the substantive requirement is the overlap $\kappa > 0$, the population analogue of the pairing fraction ρ: for an RBF k_Q it holds whenever the query *distributions* overlap, even when no individual query is shared, and it fails when models are evaluated on domains the kernel cannot bridge—where no unpaired method should be expected to compare them.

Theorem 1 (Concentration of PKPS perspectives)**.** *Under Assumptions 1–3, with at least m queries per model, for any* $\delta \in (0,1)$*, with probability at least* $1-\delta$ *there is an orthogonal* W *such that* $\max_i \|\hat{\psi}_i - W\psi_i\| \le C\kappa^{-2}\lambda_d^{-1/2}\sqrt{d}\,n\sqrt{\log(4n^2/\delta)/2m}$*, where C depends only on the kernel bounds.*

Proof sketch. Each entry of Eq. 1 is a ratio of two-sample U-statistics; Hoeffding concentration of numerator and denominator, with the denominator bounded below by κ, gives entrywise error $O(\kappa^{-2}m^{-1/2})$ (Lemma 1). A Weyl–Procrustes perturbation argument then transfers the entrywise error to the rank-d spectral embedding at the stated rate (Lemma 2). The full proof is in Appendix E. □

The rate is the paired DKPS rate with the pairing fraction ρ replaced by the population overlap κ: sampling more queries improves the representation at $m^{-1/2}$ regardless of how many queries are shared.

3.2 Inference in the PKPS

Inference is a two-step statistical problem, exactly as in the DKPS [Helm et al., 2026]: embed each model into the perspective space (Eq. 2), then regress its benchmark score on the resulting coordinate. On reference models with known scores $\{(f_i, y_i)\}$ we fit $\hat{h} : \mathbb{R}^d \to \mathscr{Y}$ on the pairs $\{(\hat{\psi}_i, y_i)\}$ and predict $\hat{y}(f) = \hat{h}(\hat{\psi}(f))$. The free choices are the bandwidth σ and the regressor $\hat{h}$.

The optimal σ depends on the degree of pairing in the data. We choose it per run by cross-validation over a median-scaled grid $\sigma \in \{0.03, \dots, 3\} \cdot \mathrm{med}$, selecting the σ whose perspective best predicts the *observed* scores under leave-one-model-out k-NN. Note that the selection criterion uses only observed scores and never the held-out scores we report. Because the grid includes near-zero bandwidths, cross-validation recovers paired DKPS when query overlap is high.

For each replicate we (i) PCA-reduce response and query embeddings on the *observed* rows only (dimension by the second Zhu–Ghodsi elbow); (ii) form D from Eq. 1; (iii) embed models by classical MDS; (iv) regress scores on the embedding, fit leave-one-family-out. The score head—logit transform, per-task standardization, two-way (model + task) bias—is identical to the low-rank matrix-completion (LRMC) baseline's. The two methods differ only in how the residual is modeled (k-NN on the perspective versus a low-rank factorization), so performance differences are attributable to the representation. We found matching the per-task standardization to be important: without it, the model offset is dominated by the tasks with the largest logit variance.

The ensemble combines the PKPS estimate with whichever score signal is available for a cell. On an *observed* cell, the score signal is the cell's own sample score, and the blend weight is a scalar fit by the same leave-one-family-out cross-validation as the regressor, against the reference models' cached full scores; the fitted weight grows with the budget, so the sample score dominates as queries accumulate. On a *missing* cell, the score signal is LRMC [Zeng and Papailiopoulos, 2026], blended with PKPS by a weight cross-validated on observed scores. No weight uses the scores of the family being predicted. (A closed-form inverse-variance weighting performs slightly worse at every budget; Appendix D.)

Query efficiency formalizes when this two-step estimator beats computing scores directly; MSE denotes mean squared error over the draw of a target model from the model population. Two further conditions are required, stated formally in Appendix E: the benchmark score is bounded and Lipschitz on the population perspective space (Assumption 4)—the working premise of any response-based predictor, visible as the smooth score gradient in Figure 1(e)—and the model population places mass on every neighborhood of the perspective space (Assumption 5), so that a target model has near reference neighbors once the cohort is large; in practice this suggests that a large, diverse reference population may be needed. Both conditions mirror Helm et al. [2026].

Definition 1 (Query efficiency; following Helm et al., 2026)**.** Fix a per-model budget m and let h_n denote a score estimator using n scored reference models. h is *query-efficient relative to* h' *at budget* m if there exists N such that $\mathrm{MSE}(h_n) \leq \mathrm{MSE}(h'_n)$ with high probability for all $n \geq N$, and *query-efficient relative to* h' if this holds at every budget at which $\mathrm{MSE}(h'_n)$ is bounded away from zero.

Theorem 2 (Query efficiency of PKPS)**.** *Fix a budget m and pairing fraction $\rho \in [0,1]$. Write PKPS $_\sigma$ for the estimator with fixed bandwidth σ, and let PKPS-CV select its bandwidth from a finite grid Σ containing the exact-matching limit $k_Q = \delta$ by validation on held-out reference scores. Under Assumptions 1–5, for any $\varepsilon > 0$:* (i) *there exist (n,m), with m polynomial in $(1/\varepsilon, 1/\kappa, n)$ and independent of ρ, at which* $\mathrm{MSE}(\textit{PKPS-CV}) \leq \varepsilon$ *with high probability;* (ii) *under a fully unpaired design (nonatomic design distributions) with non-degenerate scores, $v = \mathrm{Var}(y) > 0$, every paired-DKPS estimator has* $\mathrm{MSE} \geq v$ *at* every (n,m)*, so PKPS-CV is query-efficient relative to paired DKPS (Definition 1) with margin at least $v - 2\varepsilon$;* (iii) *for every pairing fraction ρ there exists a bandwidth $\sigma \in \Sigma$ that dominates exact matching:* $\mathrm{MSE}(\textit{PKPS}_\sigma) \leq \mathrm{MSE}(\mathrm{DKPS})$ *at every (n,m), strictly whenever any bandwidth improves on exact matching.*

Corollary 1 (Relative to the sample score)**.** *The sample score's MSE at budget m is σ_q^2/m for every n, where σ_q^2 is the per-query score variance, so by Theorem 2(i), PKPS-CV is query-efficient relative to the sample score whenever $\varepsilon < \sigma_q^2/m$, at every pairing fraction.*

Proof sketch. For (i), Theorem 1 bounds every perspective's estimation error; the nearest-neighbor error splits into two estimation terms and the population nearest-neighbor distance, which Assumption 5 makes small for n large; choosing $m = \mathrm{poly}(n, 1/\kappa, 1/\varepsilon)$ gives $\mathrm{MSE} \leq \varepsilon$, and no step involves ρ—exact matching, by contrast, retains only the ρm shared pairs, so the same argument degrades as $\rho \to 0$. For (ii), with no shared queries every identity-matched comparison is an empty sum, so a paired-DKPS prediction is a function of the reference data alone and its error is at least the score variance; (i) supplies a bandwidth attaining $\varepsilon < v$. For (iii), the δ member of the grid coincides with paired DKPS on the shared queries with the same regression head, so it attains equality; any bandwidth improving on exact matching witnesses strict domination. Cross-validation attains the best grid member up to ε (a Hoeffding oracle inequality over the finite grid), so (iii) transfers to the deployable PKPS-CV. Full proofs in Appendix E. □

Theorem 2 states that a query budget independent of the pairing fraction suffices for any target accuracy (i); that under fully unpaired designs the separation from paired DKPS is information-theoretic—with no shared queries, a paired-DKPS prediction cannot depend on the target's responses at all (ii); and that at every pairing fraction the bandwidth family contains a member dominating exact matching, since exact matching is itself in the family (iii). (i) rests on the concentration of the self-normalized product-kernel comparison, in which the population overlap κ replaces the pairing fraction ρ (Theorem 1); (ii) is an impossibility result for exact matching; and the guarantee of (iii) is made operational by cross-validation, which attains the best grid member up to ε (Appendix E). Empirically, a bridging bandwidth improves on exact matching at every $\rho < 1$ (Figure 2e).

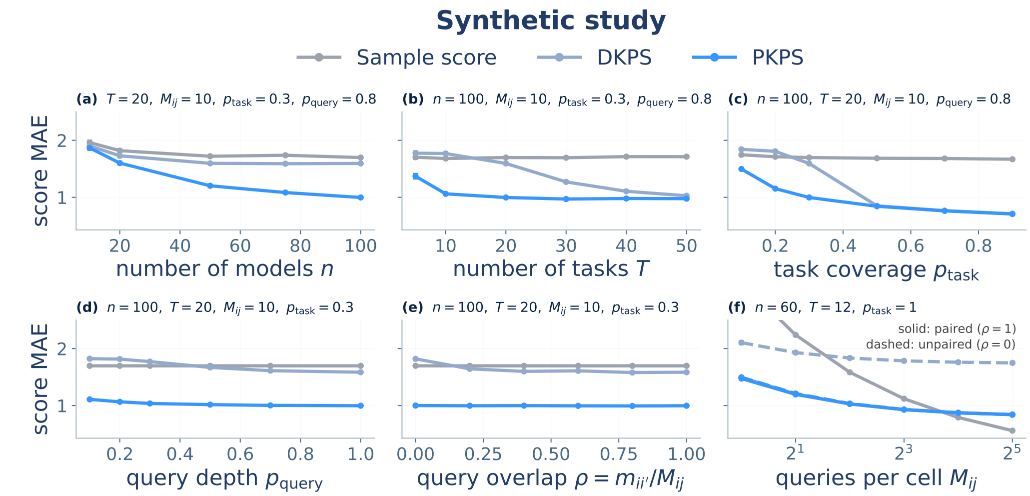

Figure 2: Score MAE on the synthetic benchmark as each lever of the observation process is varied (±1 SEM over 50 seeds; DKPS light blue, PKPS bright blue, sample score gray; fixed parameters in each panel title). **(a–e)** Completion of held-out cells as a function of cohort size n, number of tasks T, task coverage p_{task}, query depth p_{query}, and cross-model overlap $\rho = m_{ii'}/M_{ij}$; in (e), only DKPS degrades as $\rho \to 0$. **(f)** Denoising of observed cells: each cell is measured with M_{ij} noisy queries and we predict its true score, at paired (ρ=1, solid) and unpaired (ρ=0, dashed) designs; PKPS improves on the sample score at both, DKPS only when paired.

4 Experiments

We evaluate PKPS on synthetic data, where every lever of the observation process can be swept, and on two heterogeneous benchmark suites. Throughout, the two instances of Section 2 structure the experiments: query-efficient evaluation ($m_i^{(t)} > 0$) and score completion ($m_i^{(t)} = 0$).

4.1 Evaluation setup

Synthetic data. We generate n models with latent ability vectors v_i and T tasks with latent vectors t_k, scores $v_i \cdot t_k + \varepsilon$, and queries drawn near each task center. Each model answers M_{ij} queries per task, of which $m_{ii'}$ are shared with another model (overlap $\rho = m_{ii'}/M_{ij}$); a task is observed with probability p_{task} and a query with probability p_{query}.

Benchmark suites. The first suite is 18 tasks across four HELM datasets with very different response types—MATH (7 math subjects), WMT-14 (5 translation directions), MedQA (medical multiple-choice), and LegalBench (5 legal classification subsets)—on the 93 models evaluated on all four. MATH and WMT have rich free-text responses; MedQA and LegalBench responses are single tokens for which text embeddings are degenerate, so we encode each dataset's responses natively (Gemini embeddings vs. one-hot answers) in disjoint blocks. With a linear k_R this makes cross-dataset response similarity exactly zero, and a single shared query embedding with a within-domain bandwidth keeps k_Q near zero across domains, so the product kernel factorizes by domain. The second suite, from the Every Eval Ever datastore [EvalEval Coalition, 2026], is 45 modern models on 16 tasks (MATH-MC levels 1–5, GSM-MC, GPQA-Diamond, and nine judging categories from JudgeBench and RewardBench 2), with per-cell query pools drawn independently per model ($\sim$10% incidental overlap)—unpaired by construction (Appendix B).

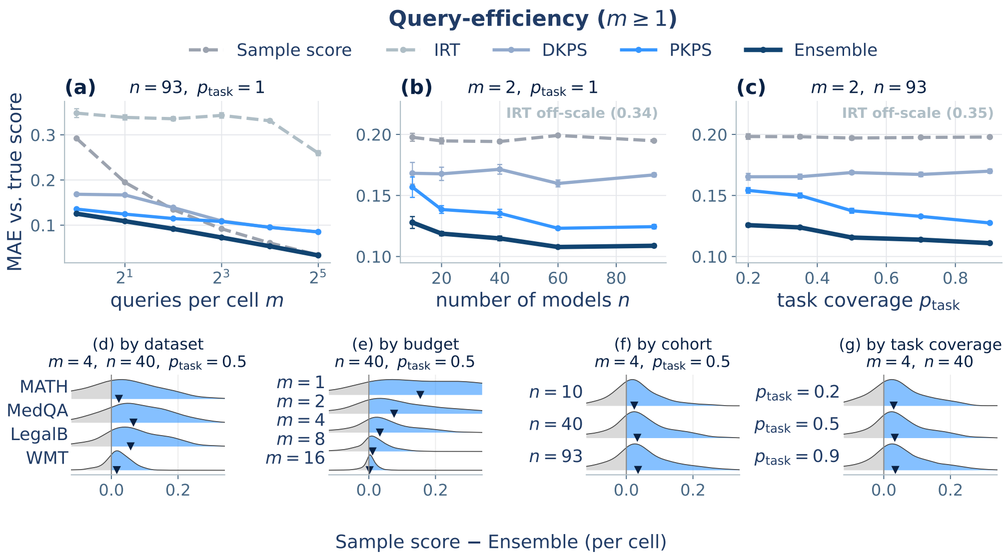

Figure 3: Query-efficient evaluation on the HELM suite. **(a–c)** MAE against the true full-evaluation score when each model answers m unpaired queries per observed cell (± 1 SEM), as a function of budget m, cohort size n, and task coverage p_{task}. IRT is poorly identified without a shared item block and is omitted from (b, c); its value is annotated. **(d–g)** Per-cell difference $\Delta =$ Sample score $-$ Ensemble MAE (red = Ensemble better) by dataset, budget, cohort, and coverage, with the other levers fixed at their sweep median; ▼ marks the per-row mean.

Prediction methods. We compare six predictors. The *sample score* is a cell's own m-query average. *IRT* fits a one-parameter logistic model on the largest shared item block [Polo et al., 2024]. *Low-rank matrix completion* (LRMC) is the logit-space imputation of Zeng and Papailiopoulos [2026] (missing cells only). *DKPS* and *PKPS* regress scores on the model embeddings of Section 3, DKPS being the identity-query-kernel limit. The *ensemble* combines PKPS with whichever score signal a cell has (Section 3).

Measurement. Every experiment predicts held-out or true cell scores and reports MAE against the full-evaluation score, mean ± 1 SEM over 16 seeds (50 on synthetic data). Sweeps vary one observation lever at a time, holding the others fixed (grids in Appendix A); all hyperparameters, including the query bandwidth, are chosen per run by the leak-free cross-validation of Section 3.

4.2 Results

PKPS outperforms DKPS on every observation lever. Figure 2 shows score MAE for PKPS, DKPS, and the sample score as each lever of the observation process is varied. PKPS outperforms DKPS at every setting, and both outperform the sample score. The margin is wide even when the data is mostly paired (a–c), since bridging similar queries, across tasks as well as within, supplies signal that exact matching does not use. The gap is largest when pairing is scarce: as p_{query} drops (d), DKPS degrades while PKPS is stable, and when ρ is swept directly (e), DKPS approaches the sample score as $\rho \to 0$ while PKPS is unaffected.

Panel (f) considers the *denoising* form of the same problem: each cell is now *observed* with M_{ij} noisy queries and we predict its true score. PKPS with a single query matches the accuracy that the sample score reaches at roughly five queries, whether or not the queries are paired; DKPS denoises only when paired (dashed). At large budgets the sample score becomes precise enough to overtake the embedding's residual bias. This is the empirical counterpart of Theorem 2.

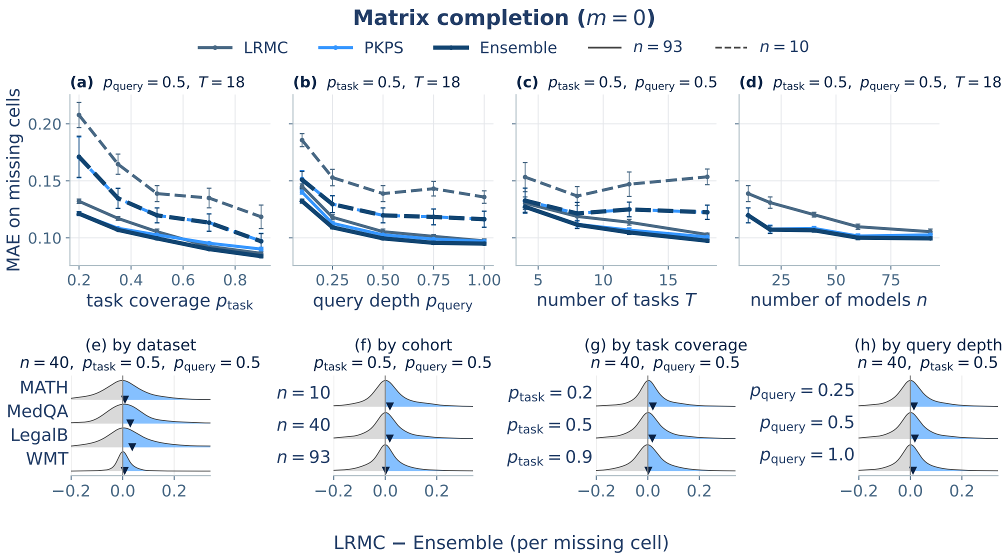

Figure 4: Score completion on the HELM suite. **(a–d)** MAE on missing cells (± 1 SEM) when a fraction p_{task} of (model, task) cells is observed at query depth p_{query}, as a function of task coverage, query depth, number of tasks, and cohort size (LRMC slate, PKPS bright blue, Ensemble navy; solid n=93, dashed n=10). **(e–h)** Per-missing-cell difference $\Delta = \text{LRMC} - \text{Ensemble}$ MAE (red = Ensemble better) by dataset, cohort, coverage, and query depth, with the other levers fixed at their sweep median; ▼ marks the per-row mean.

PKPS-based methods outperform the sample score at small query budgets. Figure 3 shows MAE as a function of the query budget m. Each model answers its own independently sampled queries (the real-data analogue of ρ=0; IRT is poorly identified in this regime) and we predict the true full-evaluation score. At small budgets the embedding-based methods are substantially more accurate than the sample score: at m=1, PKPS attains 0.136 and the ensemble 0.125 versus 0.292, so PKPS with one query matches the sample score at four. As the budget grows, the sample score eventually overtakes the embedding's residual bias (0.034 vs. 0.085 at m=32); the ensemble follows the better of its two components throughout. Panels (b, c) vary the cohort and coverage: PKPS improves steadily (0.157→0.124, 0.154→0.128; 35–37% on the second suite) while DKPS, which cannot use the additional unpaired responses, is flat.

The embedding completes missing cells where LRMC cannot. The same embedding also predicts cells that were *never run*. Figure 4 shows MAE on missing cells when a fraction p_{task} of cells is observed (each at depth p_{query}), against the LRMC baseline (the sample score is unavailable for missing cells). LRMC requires many rows to estimate its factors and degrades when models are scarce: at n=10 its MAE is 0.139 versus 0.120 for PKPS (Figure 4, dashed). At the full cohort (n=93, solid) PKPS slightly edges LRMC (0.103 vs. 0.105) and the ensemble performs best (0.099).

The per-cell gains are broad but tail-driven. The Ensemble outperforms the sample score on most cells, with the largest gains at the smallest budgets and on the rich-response datasets (Figure 3d–g); the win over LRMC is modest but stays positive in the mean across datasets, cohorts, coverage, and query depth (Figure 4e–h).

Both applications replicate on the Every Eval Ever suite. The second suite shows the same qualitative behavior with a larger completion margin (Table 1; lever sweeps in Appendix B): at m=1 PKPS attains a third of the sample score's error (0.076 vs. 0.224), matching the accuracy the sample score reaches at m=8 (0.081)—an 8× smaller budget—and on missing cells the ensemble outperforms LRMC at *every* operating point, including the full cohort (0.069 vs. 0.096 at n=45). Table 1 collects both applications on both suites; the ensemble is best or tied in every column.

The results are insensitive to representation and pipeline choices. Varying the MDS dimension over 4–24 moves PKPS's MAE by less than 0.005, and the cross-validated bandwidth matches the best fixed choice (Appendix C). Ablations of the regression head, the response kernel, the ensemble weighting rule, and the incidental query overlap in the EEE pools are in Appendix D.

5 Discussion

In this paper we introduced the PKPS, which compares models through their responses to similar, rather than identical, queries. Outside of curated leaderboards, evaluation data is typically unpaired: each model answers the queries its evaluators chose, and methods that require a shared block of identical queries do not apply. Our central finding is that benchmark prediction remains possible under unpaired evaluation: as $\rho \to 0$, DKPS and IRT degrade sharply while PKPS does not. The same embedding supports both applications we studied, denoising cheap evaluations and completing never-run cells, and the ensemble performs at least as well as either channel alone.

Limitations and future work. We close with the current limitations and the extensions they suggest. First, PKPS embeds every observed response and query with a single fixed embedder; we did not ablate this choice, and embedding functions are known to have a nonnegligible effect at small budgets [Helm et al., 2026]. Second, the block-diagonal instantiation on the HELM suite disables bridging across response domains; a shared response space that would let responses in one domain inform another remains untested. Third, computing A is quadratic in the number of queries per model pair. Fourth, Theorem 2(i) is existential (it provides no rate in n), and the strict margin of (iii) at intermediate ρ rests on a bandwidth improving on exact matching, which we establish empirically rather than by a lower bound. Finally, per-cell gains over score-only baselines are modest on dense low-rank matrices; they are decisive in the sparse, noisy, unpaired regime that motivates the problem.

Reproducibility statement

All methods are released as a documented Python package (the PKPS estimator, matrix completion, and IRT baselines), with the data-preparation, experiment, and figure scripts for both the synthetic study and the HELM suite; every experiment is seeded, all real-data numbers are means over 16 seeds (synthetic: 50), and hyperparameters are selected by the leak-free cross-validation described in Section 3 (leakage is checked by an automated test that corrupts one model's held-out scores and verifies its predictions are unchanged). Appendix A lists the suite composition, embedding models, and all hyperparameter settings.

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A Experimental details

Suite. 18 tasks over four datasets from HELM [Liang et al., 2023]: MATH (7 subjects), WMT-14 (5 translation directions), MedQA, and LegalBench (5 subsets), on the 93 models evaluated on all four; Table 2 lists every task with its query count (the loader subsamples each task to at most 120 queries with a fixed seed, so datasets contribute comparably). MATH and WMT-14 responses are free text and are embedded with `gemini-embedding-001` (3072-d); MedQA and LegalBench responses are single-token answers, encoded one-hot. Queries share a single `gemini-embedding-001` embedding space. Response and query embeddings are reduced by PCA fit on observed rows only, with dimension chosen by the second Zhu–Ghodsi elbow.

Table 2: HELM suite composition: 18 tasks over four datasets, with the number of unique queries per task. Every task is scored for all 93 shared models; the loader caps each task at 120 queries (seeded subsample).

Dataset	Task	Queries	Dataset	Task	Queries
MATH	algebra	135	WMT-14	cs–en	873
MATH	counting & probability	39	WMT-14	de–en	776
MATH	geometry	38	WMT-14	fr–en	754
MATH	intermediate algebra	52	WMT-14	hi–en	387
MATH	number theory	30	WMT-14	ru–en	613
MATH	prealgebra	86	MedQA	med_qa	1000
MATH	precalculus	57	LegalBench	abercrombie	95
LegalBench	corporate lobbying	490	LegalBench	citizenship questions	1000
LegalBench	function of decision section	367	LegalBench	proa	95

Model coverage (HELM). The 93 shared models span 20 developers: `01-ai_yi-34b`, `01-ai_yi-6b`, `01-ai_yi-large-preview`, `AlephAlpha_luminous-base`, `AlephAlpha_luminous-extended`, `AlephAlpha_luminous-supreme`, `ai21_j2-grande`, `ai21_j2-jumbo`, `ai21_jamba-1.5-large`, `ai21_jamba-1.5-mini`, `ai21_jamba-instruct`, `allenai_olmo-7b`, `amazon_nova-lite-v1`, `amazon_nova-micro-v1`, `amazon_nova-pro-v1`, `anthropic_claude-2.0`, `anthropic_claude-2.1`, `anthropic_claude-3-5-haiku-20241022`, `anthropic_claude-3-5-sonnet-20240620`, `anthropic_claude-3-5-sonnet-20241022`, `anthropic_claude-3-haiku-20240307`, `anthropic_claude-3-opus-20240229`, `anthropic_claude-3-sonnet-20240229`, `anthropic_claude-instant-1.2`, `anthropic_claude-instant-v1`, `anthropic_claude-v1.3`, `cohere_command`, `cohere_command-light`, `cohere_command-r`, `cohere_command-r-plus`, `databricks_dbrx-instruct`, `deepseek-ai_deepseek-llm-67b-chat`, `deepseek-ai_deepseek-v3`, `google_gemini-1.0-pro-001`, `google_gemini-1.0-pro-002`, `google_gemini-1.5-flash-001`, `google_gemini-1.5-flash-002`, `google_gemini-1.5-pro-001`, `google_gemini-1.5-pro-002`, `google_gemini-1.5-pro-preview-0409`, `google_gemini-2.0-flash-exp`, `google_gemma-2-27b-it`, `google_gemma-2-9b-it`, `google_gemma-7b`, `google_text-bison@001`, `google_text-unicorn@001`, `meta_llama-2-13b`, `meta_llama-2-70b`, `meta_llama-2-7b`, `meta_llama-3-70b`, `meta_llama-3-8b`, `meta_llama-3.1-405b-instruct-turbo`, `meta_llama-3.1-70b-instruct-turbo`, `meta_llama-3.1-8b-instruct-turbo`, `meta_llama-3.2-11b-vision-instruct-turbo`, `meta_llama-3.2-90b-vision-instruct-turbo`, `meta_llama-3.3-70b-instruct-turbo`, `meta_llama-65b`, `microsoft_phi-2`, `microsoft_phi-3-medium-4k-instruct`, `microsoft_phi-3-small-8k-instruct`, `mistralai_mistral-7b-instruct-v0.3`, `mistralai_mistral-7b-v0.1`, `mistralai_mistral-large-2407`, `mistralai_mistral-medium-2312`, `mistralai_mixtral-8x22b`, `mistralai_mixtral-8x7b-32kseqlen`, `mistralai_open-mistral-nemo-2407`, `nvidia_nemotron-4-340b-instruct`, `openai_gpt-3.5-turbo-0613`, `openai_gpt-4-0613`, `openai_gpt-4-1106-preview`, `openai_gpt-4-turbo-2024-04-09`, `openai_gpt-4o-2024-05-13`, `openai_gpt-4o-2024-08-06`, `openai_gpt-4o-mini-2024-07-18`, `openai_text-davinci-002`, `openai_text-davinci-003`, `qwen_qwen1.5-110b-chat`, `qwen_qwen1.5-14b`, `qwen_qwen1.5-32b`, `qwen_qwen1.5-72b`, `qwen_qwen1.5-7b`, `qwen_qwen2-72b-instruct`, `qwen_qwen2.5-72b-instruct-turbo`, `qwen_qwen2.5-7b-instruct-turbo`, `snowflake_snowflake-arctic-instruct`, `tiiuae_falcon-40b`, `tiiuae_falcon-7b`, `upstage_solar-pro-241126`, `writer_palmyra-x-004`, `writer_palmyra-x-v2`, `writer_palmyra-x-v3`.

PKPS. RBF query kernel with bandwidth selected per run by leave-one-model-out k-NN (k=8) on the observed scores over the grid $\sigma \in \{0.03, 0.1, 0.3, 1, 3\} \times$ median pairwise query distance, plus the near-delta (DKPS) limit; linear response kernel on real data, RBF on synthetic. Response blocks are capped at 48 PCA components per dataset and the shared query space at 64; each (model, query) pair uses one sampled response (r=1). Models are embedded by classical MDS with d=8 throughout, following Helm et al. [2026] and fixed a priori (d = latent dimension on synthetic). Appendix C shows the results are insensitive to d over 4–24, and that the cross-validated bandwidth tracks the best fixed choice.

Prediction head. All embedding methods share one head: logit-transformed scores, per-task standardization to unit variance, a two-way (model + task) additive bias, and a distance-weighted k-NN (k=5) regression of the residual on the model embedding, fit *leave-one-family-out*: all models from the target's developer are excluded from its training set, so predictions never lean on near-duplicate siblings [Helm et al., 2026]. Held-out cells are predicted directly; observed cells leave their own target out.

Baselines. Low-rank matrix completion (LRMC) follows BenchPress [Zeng and Papailiopoulos, 2026]: logit space, per-task standardization, two-way bias, rank-2 ALS with ridge λ=0.1, averaged over 5 random initializations; on observed cells it is cross-fit so a cell's prediction never sees its own sample. IRT is a 1PL (Rasch) model on binary tasks: item difficulties by maximum likelihood (L-BFGS) on the largest fully-observed (model $\times$ item) block, per-model ability by bounded scalar MLE, clipped to $[-4,4]$. The sample baseline is the cell's own m-query mean.

Ensemble. On observed cells, the prediction is $\alpha\,\hat{y}_{\text{sample}} + (1-\alpha)\,\hat{y}_{PKPS}$ with a scalar α per held-out family, selected on a 21-point grid by leave-one-family-out validation against the other families' cached full scores (the same information the regressor trains on); on missing cells, PKPS and LRMC are blended with a weight fit by held-out cross-validation against observed scores. No quantity ever uses the scores of the family being predicted. Appendix D compares this rule to a closed-form inverse-variance weighting and to the fixed m/M schedule of Helm et al. [2026].

Sweeps and reporting. All real-data sweeps use 16 seeds; curves and table entries report the per-seed mean over tasks, then the mean ($\pm$1 SEM) over seeds. Query efficiency: budgets $m \in \{1,2,4,8,16,32\}$ at full cohort and coverage; cohort $n \in \{10,20,40,60,93\}$ (HELM) or $\{10,20,30,45\}$ (EEE) and task coverage $p_{\text{task}} \in \{0.2,0.35,0.5,0.7,0.9\}$, both at the few-query point m=2. Completion: task coverage as above and query depth $p_{\text{query}} \in \{0.1,0.25,0.5,0.75,1.0\}$ with cohort lines $n \in \{10,\text{full}\}$; task count $T \in \{4,8,12,18\}$ (HELM) or $\{4,8,12,16\}$ (EEE); the cohort sweep holds p_{task}=p_{query}=0.5.

Synthetic study. 50 seeds per configuration; latent dimension 5, observed embedding dimension 20, score noise 0.1, response noise 0.5; both kernels RBF; k-NN regression with k=5. Panel-specific settings are given in each figure title.

Compute. All experiments run on a single multi-core CPU node; text embeddings are precomputed once through the embedding API and cached. Embedding the EEE suite (30,304 pooled responses plus 7,665 queries, ∼11M tokens) took ∼13 minutes and under \$2 of API credit; each 16-seed sweep completes in minutes (EEE) to hours (HELM) on the same node.

B The Every Eval Ever suite

The second suite is built from the Every Eval Ever datastore [EvalEval Coalition, 2026], an MIT-licensed, community-maintained collection of item-level evaluation results in a unified format (inputs, raw model outputs, and per-item scores). We take the five benchmarks with the widest

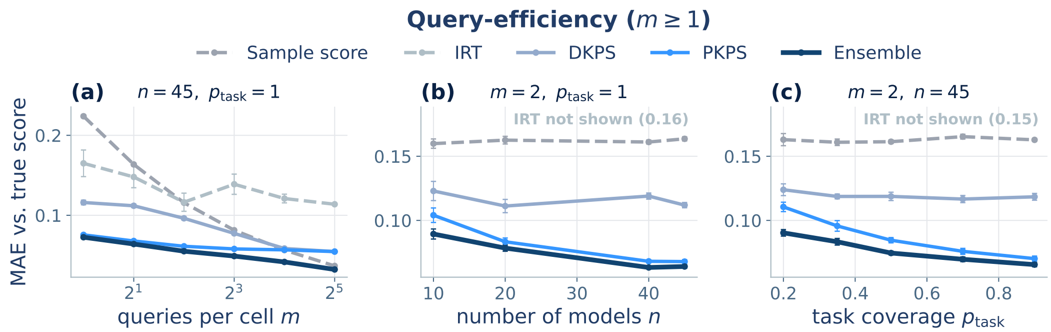

Figure 5: Query-efficient evaluation on the Every Eval Ever suite (45 models, 16 tasks; MAE against the true full-evaluation score, ± 1 SEM over 16 seeds). The protocol matches Figure 3. PKPS and the Ensemble are the most accurate at every budget; DKPS trails because the query pools are unpaired, and IRT is poorly identified.

item-level model coverage—MATH-MC, GSM-MC, GPQA-Diamond, JudgeBench, and Reward-Bench 2—restricted to the 45 models present on all five; sub-tasks (MATH-MC levels 1–5; four JudgeBench and five RewardBench 2 categories) give 16 tasks (Table 3). Every domain is free text (including the judging benchmarks' rationales), so all responses and queries are embedded with `gemini-embedding-001` and blocked by benchmark exactly as in Appendix A. Each cell's observable pool is 32 queries drawn with a model-dependent seed, so two models share only incidental queries ($\sim 10\%$); the query-efficiency budget m and the completion depth p_{query} subsample this pool, and the prediction target is always the cell's full-depth mean over all items, which is never embedded. All scores are binary, so IRT applies to every task (on the HELM suite it excludes WMT). Figures 5 and 6 repeat the lever sweeps of Figures 3 and 4 on this suite; Table 1 summarizes.

Table 3: EEE suite composition: 16 tasks over five benchmarks, with the number of items per (model, task) cell. Every task is fully answered by all 45 shared models; each cell's observable pool is 32 items, sampled per model.

Benchmark	Task	Items	Benchmark	Task	Items
MATH-MC	level 1	430	JudgeBench	coding	42
MATH-MC	level 2	882	JudgeBench	knowledge	154
MATH-MC	level 3	1115	JudgeBench	math	56
MATH-MC	level 4	1199	JudgeBench	reasoning	98
MATH-MC	level 5	1288	RewardBench 2	factuality	475
GSM-MC	gsm_mc	1319	RewardBench 2	focus	495
GPQA-Diamond	gpqa_diamond	198	RewardBench 2	math	183
RewardBench 2	precise if	160	RewardBench 2	safety	450

Model coverage (EEE). The 45 models shared by all five benchmarks span 13 developers: `cohere/c4ai-command-a-03-2025`, `cohere/c4ai-command-r-08-2024`, `cohere/c4ai-command-r-plus-08-2024`, `cohere/c4ai-command-r7b-12-2024`, `cohere/command-a-reasoning-08-2025`, `deepseek/deepseek-r1-0528`, `deepseek/deepseek-v3-1-terminus`, `deepseek/deepseek-v3-2`, `deepseek/deepseek-v3-2-speciale`, `deepseek/deepseek-v4-flash-fp8`, `deepseek/deepseek-v4-pro`, `google/gemini-3-1-pro-preview`, `google/gemma-2-27b-it`, `google/gemma-2-9b-it`, `google/gemma-3-12b-it`, `google/gemma-3-27b-it`, `google/gemma-4-e2b-it`, `google/gemma-4-e4b-it`, `llm360/k2-v2-instruct`, `meta/llama-4-maverick-17b-128e-instruct-fp8`, `meta/meta-llama-3-1-8b-instruct`, `minimax/minimax-m2-5`, `mistral/mistral-small-4-119b-2603`, `moonshot/kimi-k2-5`, `moonshot/kimi-k2-6`, `moonshot/kimi-k2-thinking`, `openai/gpt-5-5`, `openai/gpt-5-mini`, `openai/gpt-5-nano`, `openai/gpt-oss-120b`, `openai/gpt-oss-20b`, `qwen/qwen3-30b-a3b-thinking-2507`, `qwen/qwen3-5-122b-a10b`, `qwen/qwen3-5-2b`, `qwen/qwen3-5-397b-a17b`, `qwen/qwen3-6-35b-a3b`, `qwen/qwen3-6-plus`, `qwen/qwen3-next-80b-a3b-thinking`, `stepfun/step-3-5-flash`, `xiaomi/mimo-v2-flash`, `zai-org/glm-4-6-fp8`, `zai-org/glm-4-7-flash`, `zai-org/glm-4-7-fp8`, `zai-org/glm-5-1-fp8`, `zai-org/glm-5-fp8`.

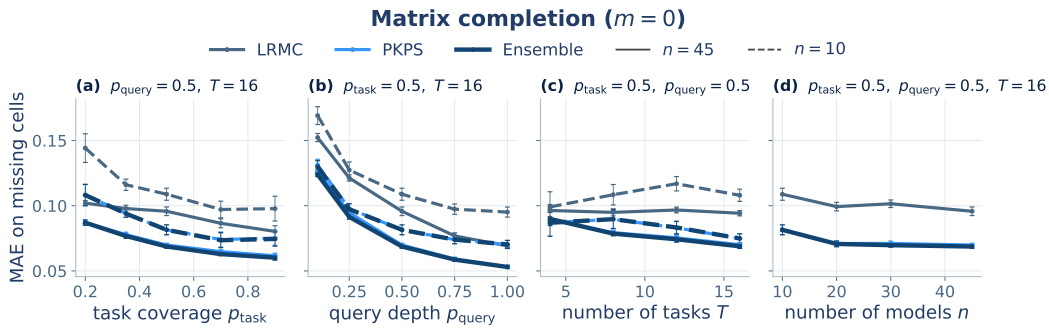

Figure 6: Matrix completion on the Every Eval Ever suite (MAE on missing cells, ± 1 SEM; solid n=45, dashed n=10). The protocol matches Figure 4. Unlike on the HELM suite, PKPS and the Ensemble outperform LRMC at every operating point.

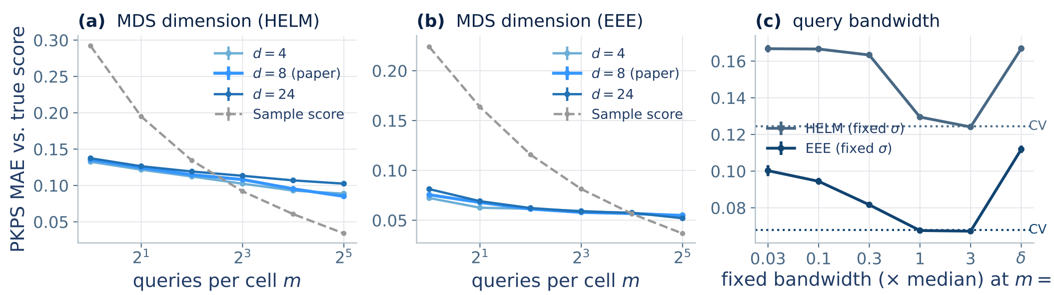

Figure 7: Sensitivity analyses on the query-efficiency experiment (± 1 SEM, 16 seeds). **(a, b)** PKPS MAE against budget at MDS dimension $d \in \{4, 8, 24\}$ (paper choice in red; sample dashed); the curves are nearly indistinguishable on both suites. **(c)** MAE at m=2 with the query bandwidth fixed at each grid multiplier (δ = exact matching); dotted lines mark the cross-validated selection, which matches the best fixed choice on both suites.

C Sensitivity analyses

Figure 7 probes the two representation hyperparameters on the query-efficiency experiment (16 seeds throughout). **MDS dimension (a, b).** Repeating the budget sweep at $d \in \{4, 8, 24\}$ moves PKPS's MAE by less than 0.005 at every budget on both suites (at the smallest budgets, smaller d is very slightly better—fewer coordinates to estimate from one noisy query per cell); the paper's choice (d=8, following Helm et al., 2026) is never the worst point at any budget. **Query bandwidth (c).** Fixing the bandwidth at each grid multiplier instead of cross-validating traces the expected curve: near-delta multipliers and the exact-matching δ limit are substantially worse (unpaired pools leave them little to compare), broad multipliers are best, and the leak-free cross-validated choice matches the best fixed multiplier on both suites (0.126 vs. 0.125 on HELM, 0.069 vs. 0.068 on EEE)—the one hyperparameter that matters is the one the method selects automatically. Sensitivity to the number of reference models is part of the main experiments (the cohort sweeps of Figures 3b and 4d).

D Pipeline ablations

Each ablation repeats the query-efficiency budget sweep (16 seeds, both suites) with one pipeline choice changed; all other settings match Appendix A.

Ensemble weighting. We compare the paper's cross-validated scalar weight against two alternatives: a closed-form inverse-variance rule (the sample score weighted by $e^2_{PKPS}/(e^2_{PKPS}+\tilde{p}(1-\tilde{p})/m)$ per cell, with $\tilde{p}$ the PKPS estimate and e^2_{PKPS} the depth-weighted, noise-debiased disagreement with the observed samples) and the fixed schedule $\alpha = m/M$ of Helm et al. [2026]. The cross-validated weight is best at every budget on both suites: on HELM it improves on the inverse-variance rule from 0.132 to 0.125 at m=1 and from 0.077 to 0.073 at m=8; on EEE from 0.084 to 0.073 and from 0.052 to 0.049. The m/M schedule trails both on HELM (0.096 at m=8) and is close to the cross-validated weight on EEE, where the small pools (M=32) let m/M reach 1.

Regression head. Replacing the distance-weighted k-NN head with ordinary least squares helps slightly on HELM (0.132 vs. 0.136 at m=1; 0.080 vs. 0.085 at m=32), consistent with Helm et al. [2026], but hurts on EEE, increasingly so at large budgets (0.066 vs. 0.055 at m=32). We use k-NN as the robust default across suites.

Response kernel. An RBF response kernel in place of the linear one changes PKPS's MAE by at most 0.003 at every budget on both suites; the linear choice is justified by the block-diagonal factorization (Section 4.1) at no accuracy cost.

Incidental overlap. The EEE pools share ∼10% of items across models by chance. Enforcing exactly disjoint query sets (true $\rho = 0$) changes PKPS by at most 0.005 at $m \leq 8$ (0.080 vs. 0.076 at m=1), and IRT becomes inapplicable outright—no shared item block exists. At $m \geq 16$ the disjointness constraint exhausts the 32-item pools and mechanically shortens the sample score's effective depth, so comparisons there reflect the constraint rather than the methods.

E Proofs of Theorems 1 and 2 and Corollary 1

Formal definitions and assumptions. We collect the formal statements referenced in Section 3.

Definition 2 (Population PKPS)**.** Let each model i's (query, response) pairs be drawn i.i.d. from a design distribution P_i. The *population comparison* is $A^*_{ii'} = \mathbb{E}[k_Q(u,u')\,k_R(x_i,x_{i'})]\,/\,\mathbb{E}[k_Q(u,u')]$ with draws from P_i and $P_{i'}$ independent, the population squared distances are $D^{*2}(i,i') = A^*_{ii} + A^*_{i'i'} - 2A^*_{ii'}$, and the *population perspectives* $\psi_1,\ldots,\psi_n \in \mathbb{R}^d$ are the rank-d spectral embedding of the centered Gram matrix B^* of D^{*2}.

Assumption 1 (Boundedness)**.** Kernels and embedded responses are bounded.

Assumption 2 (Population overlap)**.** The *cross-model query-kernel mass* is bounded below: $\kappa = \inf_{i,i'} \mathbb{E}[k_Q(u,u')] > 0$.

Assumption 3 (Spectral identifiability)**.** B^* has rank d with spectral gap $\lambda_d > 0$.

Assumption 4 (Score regularity)**.** The benchmark score is bounded and γ-Lipschitz on the population perspective space.

Assumption 5 (Model support)**.** The model distribution places mass on every neighborhood of the population perspective space.

Population quantities. Queries for model i are drawn i.i.d. from a design distribution over embedded queries; write $z = (u, x_i(u))$ for an embedded query–response pair (response noise included) and P_i for its law. By Assumption 1 the kernels are bounded: $k_Q \in [0,1]$ after normalization (RBF) and $|k_R| \leq B_R$. Define the population comparison

$$A^*_{ii'} \;=\; \frac{\mathbb{E}\left[k_Q(u,u')\,k_R(x_i(u),x_{i'}(u'))\right]}{\mathbb{E}\left[k_Q(u,u')\right]}, \qquad z \sim P_i,\; z' \sim P_{i'} \text{ independent},$$

population squared distances $D^{*2}(i,i') = A^*_{ii} + A^*_{i'i'} - 2A^*_{ii'}$, the centered Gram matrix $B^* = -\frac{1}{2}JD^{*2}J$ with $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, and population perspectives $\Psi^* \in \mathbb{R}^{n\times d}$ its rank-d spectral embedding, which exists with $\lambda_d(B^*) > 0$ by Assumption 3 (cf. Definition 2).

Lemma 1 (Affinity concentration)**.** *Under Assumptions 1 and 2, for models i,i' with $\min(|Q_i|,|Q_{i'}|) \geq m$ and any $\delta \in (0,1)$: with probability at least $1-\delta$,*

$$|A_{ii'} - A^*_{ii'}| \;\leq\; \frac{2(1+B_R)}{\kappa^2}\, t_m(\delta), \qquad t_m(\delta) = \bar{B}\sqrt{\frac{\log(4/\delta)}{2m}}, \quad \bar{B} = \max(2B_R, 1),$$

provided $t_m(\delta) \leq \kappa/2$.

Proof. The numerator $N = \frac{1}{|Q_i||Q_{i'}|}\sum_{j,l} k_Q(u_j,u'_l)\,k_R(x_{ij},x_{i'l})$ and denominator $Z = \frac{1}{|Q_i||Q_{i'}|}\sum_{j,l} k_Q(u_j,u'_l)$ of Eq. 1 are two-sample U-statistics of order $(1,1)$ in the i.i.d. pairs $z_j \sim P_i$, $z'_l \sim P_{i'}$, with kernels of range at most $2B_R$ and 1 respectively (response noise enters only through the bounded kernel argument, so the number of responses per query may stay fixed). By Hoeffding's inequality for two-sample U-statistics [Hoeffding, 1963, Section 5], each deviates from its mean by more than its range times $\sqrt{\log(4/\delta)/2m}$ with probability at most $\delta/2$; since $\bar{B}$ dominates both ranges, a union bound gives $|N-N^*| \leq t_m(\delta)$ and $|Z-Z^*| \leq t_m(\delta)$ jointly with probability $\geq 1-\delta$. On this event, since $Z^* \geq \kappa$ by Assumption 2 and $t_m(\delta) \leq \kappa/2$, we have $Z \geq \kappa/2$, and

$$|A_{ii'} - A^*_{ii'}| = \frac{|NZ^* - N^*Z|}{ZZ^*} \;\leq\; \frac{|N-N^*|}{Z} + \frac{|N^*|\,|Z-Z^*|}{ZZ^*} \;\leq\; \frac{2t_m(\delta)}{\kappa} + \frac{2B_R\,t_m(\delta)}{\kappa^2} \;\leq\; \frac{2(1+B_R)}{\kappa^2}\,t_m(\delta),$$

using $|N^*| \leq B_R$ and $\kappa \leq 1$.

□

Lemma 2 (Perspective perturbation)**.** *Let $\varepsilon_\infty = \max_{i,i'} |A_{ii'} - A^*_{ii'}|$. There is an orthogonal $W \in \mathbb{R}^{d\times d}$ such that*

$$\max_i \|\hat{\psi}_i - W\psi_i\| \;\leq\; \frac{25\sqrt{d}\,n\,\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}\,.$$

Proof. Entrywise, $|D^2 - D^{*2}| \leq 4\varepsilon_\infty$, and double centering inflates an entrywise bound by at most a factor of 4, so $E = B - B^*$ satisfies $\|E\|_{\text{op}} \leq \|E\|_F \leq 8\,n\,\varepsilon_\infty$. Let B_d be the top-d truncation of B that classical MDS embeds ($B_d = \hat{\Psi}\hat{\Psi}^\top$, positive semidefinite). By Weyl's inequality every eigenvalue of B beyond the d-th, and every negative eigenvalue, has magnitude at most $\|E\|_{\text{op}}$ (since $\lambda_{d+1}(B^*) = 0$ and $B^* \succeq 0$), so $\|B_d - B\|_{\text{op}} \leq \|E\|_{\text{op}}$. The matrix $B_d - B^*$ has rank at most $2d$, hence

$$\|B_d - B^*\|_F \;\leq\; \sqrt{2d}\,\|B_d - B^*\|_{\text{op}} \;\leq\; \sqrt{2d}\left(\|B_d - B\|_{\text{op}} + \|E\|_{\text{op}}\right) \;\leq\; 2\sqrt{2d}\,\|E\|_F.$$

Both $B_d = \hat{\Psi}\hat{\Psi}^\top$ and $B^* = \Psi^*\Psi^{*\top}$ are positive semidefinite of rank at most d, so by the Procrustes alignment lemma of Tu et al. [2016, Lemma 5.4],

$$\min_{W\in O(d)} \|\hat{\Psi} - \Psi^* W\|_F^2 \;\leq\; \frac{\|B_d - B^*\|_F^2}{2(\sqrt{2}-1)\,\lambda_d(B^*)}\,.$$

Combining, $\max_i \|\hat{\psi}_i - W\psi_i\| \leq \|\hat{\Psi} - \Psi^* W\|_F \leq \frac{2\sqrt{2d}}{\sqrt{2(\sqrt{2}-1)}} \cdot \frac{8n\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}} \leq \frac{25\sqrt{d}\,n\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}$. Nearest-neighbor distances are invariant to W. □

Proof of Theorem 2(i). Take scores in $[0,1]$ without loss of generality (Assumption 4). Fix $\varepsilon > 0$ and set targets $c \le \sqrt{\varepsilon/2}/(8\gamma)$ for the estimation error and $\varepsilon' \le \sqrt{\varepsilon/2}/(2\gamma)$ for the population nearest-neighbor distance. By Assumption 5 there is $n_0(\varepsilon')$ such that for $n \ge n_0$, a reference model within population distance ε' of the target exists except with probability at most $\varepsilon/8$. Fix such an n. Applying Lemma 1 with confidence $\delta = \varepsilon/(8n^2)$ to every pair, a union bound and Lemma 2 give $\max_i \|\hat{\psi}_i - W\psi_i\| \le c$ except with probability at most $\varepsilon/8$, once

$$m \;\ge\; C' \,\frac{d\,(1+B_R)^2\,\bar{B}^2\,n^2\log(n/\varepsilon)}{\kappa^4\,\lambda_d\,c^2} \;=\; \mathrm{poly}(n, 1/\kappa, 1/\varepsilon),$$

for an absolute constant C' (this choice also satisfies the proviso $t_m \le \kappa/2$ of Lemma 1), and does not involve the pairing fraction ρ. On the intersection of the two events, for a target model f with nearest estimated reference neighbor f^* (perspectives $\psi^*, \hat{\psi}^*$; the target's $\psi, \hat{\psi}$),

$$|\hat{y}(f) - y(f)| \;\le\; \gamma\|\psi^* - \psi\| \;\le\; \gamma\big(\|\hat{\psi}^* - W\psi^*\| + \|\hat{\psi}^* - \hat{\psi}\| + \|\hat{\psi} - W\psi\|\big) \;\le\; \gamma(2c + \hat{\delta}),$$

using Assumption 4 and orthogonal invariance. The estimated nearest-neighbor distance $\hat{\delta}$ is at most the estimated distance to the ε'-close reference, itself at most $\varepsilon' + 2c$ by the same alignment argument, so $|\hat{y} - y| \le \gamma(4c + \varepsilon') \le \sqrt{\varepsilon/2}$. Squaring, and adding the two failure events (squared errors are bounded by 1), $\mathrm{MSE}(\hat{y}) \le \varepsilon/2 + \varepsilon/4 \le \varepsilon$. □

Relative to the sample score. The sample score at budget m is the mean of m i.i.d. per-query scores and uses no reference models, so its MSE is σ_q^2/m for every n, where σ_q^2 is the per-query score variance ($\le 1/4$ for scores in $[0,1]$). For any $\varepsilon < \sigma_q^2/m$ at the budget of Theorem 2(i), the estimator satisfies $\mathrm{MSE} \le \varepsilon < \sigma_q^2/m$ for all sufficiently large n: query efficiency relative to the sample score in the sense of Definition 1, proving Corollary 1.

Contrast with paired DKPS. With $k_Q = \delta$, only identically-shared query pairs enter the double sum, so the U-statistic runs over the ρm shared queries and the rate in Lemma 1 becomes $O_P\big((\rho m)^{-1/2}\big)$: DKPS's guarantee requires $\rho m \to \infty$ and vacates in the unpaired limit $\rho \to 0$, where its distance is undefined (Section 3).

Proofs of Theorem 2(ii) and (iii). For (iii), existence is witnessed by the exact-matching member: the $k_Q = \delta$ estimator coincides with paired DKPS on the shared queries with the same regression head (Section 3.1), so $\sigma = \delta$ attains $\mathrm{MSE}(PKPS_\sigma) = \mathrm{MSE}(\mathrm{DKPS})$ at every (n, m), and any bandwidth improving on exact matching witnesses strict domination. The remainder establishes the operational form: cross-validation attains the best grid member up to ε. Split the reference models into a training set T and a validation set V with $|V| = n_v$. For each $\sigma \in \Sigma$, build the perspective space on T, embed any model outside T out of sample by its Eq. 1 comparison row to T (the standard out-of-sample MDS coordinate), and let $\hat{y}_\sigma$ be nearest-neighbor regression on T's (perspective, score) pairs. The selected bandwidth is $\hat{\sigma} = \arg\min_{\sigma\in\Sigma} \hat{L}(\sigma)$ with $\hat{L}(\sigma) = \frac{1}{n_v}\sum_{i\in V}(\hat{y}_\sigma(f_i) - y_i)^2$. Conditional on T, the validation losses are i.i.d. across the draw of validation models and bounded in $[0,1]$, so by Hoeffding's inequality and a union bound over Σ, with probability at least $1 - \delta'$, $\sup_{\sigma\in\Sigma} |\hat{L}(\sigma) - L(\sigma)| \le \tau = \sqrt{\log(2|\Sigma|/\delta')/2n_v}$, where $L(\sigma)$ is the MSE on a fresh target. On this event,

$$L(\hat{\sigma}) \;\le\; \hat{L}(\hat{\sigma}) + \tau \;\le\; \hat{L}(\delta) + \tau \;\le\; L(\delta) + 2\tau \;=\; \mathrm{MSE}(\mathrm{DKPS}) + 2\tau,$$

since the $k_Q = \delta$ member of the family coincides with paired DKPS on the shared queries with the same regression head. Taking n_v large enough that $2\tau \le \varepsilon$ transfers the guarantee of (iii) to PKPS-CV at every (m, ρ). If some $\sigma^\star$ has $\Delta = L(\delta) - L(\sigma^\star) > 0$, the same event gives $L(\hat{\sigma}) \le L(\sigma^\star) + 2\tau = \mathrm{MSE}(\mathrm{DKPS}) - \Delta + 2\tau$, strict for $2\tau < \Delta$. Finally, at $\rho = 0$ with nonatomic designs, $\Pr(q_j = q'_l) = 0$ for every pair of independently drawn queries, so almost surely every identity-matched sum is empty: any measurable completion of the DKPS member predicts as a function of the reference data alone, independent of the target, whence $\mathrm{MSE}(\mathrm{DKPS}) = v + \mathbb{E}[(\mathbb{E}y - \hat{y})^2] \geq v$, while under Assumptions 1–5 part (i) supplies a bandwidth attaining $\mathrm{MSE} \leq \varepsilon$, so $\Delta \geq v - \varepsilon$ and the margin is at least $v - \varepsilon - 2\tau \geq v - 2\varepsilon$. In practice we use leave-one-model-out validation and a joint embedding (Section 3); the split protocol is analyzed for independence of the validation losses. $\square$